MSG No. 181.174 $P6_422'$ [Type II, hexagonal]

Table 1: Wyckoff site: `3a`, site symmetry: `2221'`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,4,7,11,13,16,19,23]`
2	$\left[0,\ 0,\ \frac{2}{3}\right]$	`[2,5,9,10,14,17,21,22]`
3	$\left[0,\ 0,\ \frac{1}{3}\right]$	`[3,6,8,12,15,18,20,24]`

Table 2: Wyckoff site: `3b`, site symmetry: `2221'`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[1,4,7,11,13,16,19,23]`
2	$\left[0,\ 0,\ \frac{1}{6}\right]$	`[2,5,9,10,14,17,21,22]`
3	$\left[0,\ 0,\ \frac{5}{6}\right]$	`[3,6,8,12,15,18,20,24]`

Table 3: Wyckoff site: `3c`, site symmetry: `2221'`

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[1,4,7,11,13,16,19,23]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{2}{3}\right]$	`[2,5,9,10,14,17,21,22]`
3	$\left[0,\ \frac{1}{2},\ \frac{1}{3}\right]$	`[3,6,8,12,15,18,20,24]`

Table 4: Wyckoff site: `3d`, site symmetry: `2221'`

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[1,4,7,11,13,16,19,23]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{6}\right]$	`[2,5,9,10,14,17,21,22]`
3	$\left[0,\ \frac{1}{2},\ \frac{5}{6}\right]$	`[3,6,8,12,15,18,20,24]`

Table 5: Wyckoff site: `6e`, site symmetry: `2..1'`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,4,13,16]`
2	$\left[0,\ 0,\ z+\frac{2}{3}\right]$	`[2,5,14,17]`
3	$\left[0,\ 0,\ z+\frac{1}{3}\right]$	`[3,6,15,18]`
4	$[0,\ 0,\ -z]$	`[7,11,19,23]`
5	$\left[0,\ 0,\ \frac{1}{3}-z\right]$	`[8,12,20,24]`
6	$\left[0,\ 0,\ \frac{2}{3}-z\right]$	`[9,10,21,22]`

Table 6: Wyckoff site: `6f`, site symmetry: `2..1'`

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ z\right]$	`[1,4,13,16]`

continued ...

Table 6

No.	position	mapping
2	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{2}{3}\right]$	[2,5,14,17]
3	$\left[0,\ \frac{1}{2},\ z+\frac{1}{3}\right]$	[3,6,15,18]
4	$\left[\frac{1}{2},\ 0,\ -z\right]$	[7,11,19,23]
5	$\left[0,\ \frac{1}{2},\ \frac{1}{3}-z\right]$	[8,12,20,24]
6	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{2}{3}-z\right]$	[9,10,21,22]

Table 7: Wyckoff site: 6g, site symmetry: .2.1'

No.	position	mapping
1	$[x,\ 0,\ 0]$	[1,7,13,19]
2	$\left[x,\ x,\ \frac{2}{3}\right]$	[2,10,14,22]
3	$\left[0,\ x,\ \frac{1}{3}\right]$	[3,8,15,20]
4	$[-x,\ 0,\ 0]$	[4,11,16,23]
5	$\left[-x,\ -x,\ \frac{2}{3}\right]$	[5,9,17,21]
6	$\left[0,\ -x,\ \frac{1}{3}\right]$	[6,12,18,24]

Table 8: Wyckoff site: 6h, site symmetry: .2.1'

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{2}\right]$	[1,7,13,19]
2	$\left[x,\ x,\ \frac{1}{6}\right]$	[2,10,14,22]
3	$\left[0,\ x,\ \frac{5}{6}\right]$	[3,8,15,20]
4	$\left[-x,\ 0,\ \frac{1}{2}\right]$	[4,11,16,23]
5	$\left[-x,\ -x,\ \frac{1}{6}\right]$	[5,9,17,21]
6	$\left[0,\ -x,\ \frac{5}{6}\right]$	[6,12,18,24]

Table 9: Wyckoff site: 6i, site symmetry: ..21'

No.	position	mapping
1	$[x,\ 2x,\ 0]$	[1,11,13,23]
2	$\left[-x,\ x,\ \frac{2}{3}\right]$	[2,9,14,21]
3	$\left[-2x,\ -x,\ \frac{1}{3}\right]$	[3,12,15,24]
4	$[-x,\ -2x,\ 0]$	[4,7,16,19]
5	$\left[x,\ -x,\ \frac{2}{3}\right]$	[5,10,17,22]
6	$\left[2x,\ x,\ \frac{1}{3}\right]$	[6,8,18,20]

Table 10: Wyckoff site: 6j, site symmetry: ..21'

No.	position	mapping
1	$\left[x,\ 2x,\ \frac{1}{2}\right]$	[1,11,13,23]
2	$\left[-x,\ x,\ \frac{1}{6}\right]$	[2,9,14,21]
3	$\left[-2x,\ -x,\ \frac{5}{6}\right]$	[3,12,15,24]

continued ...

Table 10

No.	position	mapping
4	$\left[-x,\, -2x,\, \frac{1}{2}\right]$	[4,7,16,19]
5	$\left[x,\, -x,\, \frac{1}{6}\right]$	[5,10,17,22]
6	$\left[2x,\, x,\, \frac{5}{6}\right]$	[6,8,18,20]

Table 11: Wyckoff site: 12k, site symmetry: 11'

No.	position	mapping
1	$[x,\, y,\, z]$	[1,13]
2	$\left[x-y,\, x,\, z+\frac{2}{3}\right]$	[2,14]
3	$\left[-y,\, x-y,\, z+\frac{1}{3}\right]$	[3,15]
4	$[-x,\, -y,\, z]$	[4,16]
5	$\left[-x+y,\, -x,\, z+\frac{2}{3}\right]$	[5,17]
6	$\left[y,\, -x+y,\, z+\frac{1}{3}\right]$	[6,18]
7	$[x-y,\, -y,\, -z]$	[7,19]
8	$\left[y,\, x,\, \frac{1}{3}-z\right]$	[8,20]
9	$\left[-x,\, -x+y,\, \frac{2}{3}-z\right]$	[9,21]
10	$\left[x,\, x-y,\, \frac{2}{3}-z\right]$	[10,22]
11	$[-x+y,\, y,\, -z]$	[11,23]
12	$\left[-y,\, -x,\, \frac{1}{3}-z\right]$	[12,24]